

## Code-Layout, Readability and Reusability

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 03 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-9794                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                    | Followed (Yes / No / Partial) | Remarks                                                      |
|------|---------------------------|------------------------------------------------|-------------------------------|--------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code. | Yes                           | Proper indentation maintained                                |
| 2    | Proper File Structure     | Files and folders are logically organized      | Yes                           | Separate folders for dataset, model, templates, static files |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly        | Yes                           | Descriptive variable names used                              |
| 4    | Function / Method Names   | Functions are descriptively named              | Yes                           | Easy to understand and maintain                              |
| 5    | Code Comments             | Inline and block comments explain logic        | Yes                           | Important sections documented                                |
| 6    | Modular Design            | Code is split into reusable functions/modules  | Yes                           | Separate preprocessing, training, prediction modules         |
| 7    | No Redundant Code         | Duplicate or unused code is removed            | Yes                           | Optimized implementation                                     |
| 8    | Error Handling            | Exceptions and errors are handled gracefully   | Yes                           | Input validation and exception handling included             |

### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technology | Where Reused | Reusability Level (High / Medium / Low) |
|------|-------------------------|-----------------------|--------------|-----------------------------------------|
|------|-------------------------|-----------------------|--------------|-----------------------------------------|



|   |                       |                |                        |      |
|---|-----------------------|----------------|------------------------|------|
| 1 |                       | Python, Pandas | Data Cleaning          | High |
| 2 | Feature Encoding      | Scikit-learn   | Data Preprocessing     | High |
| 3 | Feature Scaling       | Scikit-learn   | Prediction             | High |
| 4 | ML Prediction Model   | Scikit-learn   | Credit Card Prediction | High |
| 5 | Flask Web Application | Flask          | User Interface         | High |
| 6 | HTML Templates        | HTML/CSS       | Web Pages              | High |

| Aspect                  | Rating   | Comments                              |
|-------------------------|----------|---------------------------------------|
| Code Layout & Structure | 5        | Well-organized project structure      |
| Readability             | 5        | Easy to understand and maintain       |
| Reusability             | 5        | Reusable ML modules                   |
| Documentation           | 5        | Proper comments provided              |
| Overall Score           | 5/5/2026 | High-quality and maintainable project |